What is a sensor?

IoT · Level 1 · Lesson 1



Today we will

- Explain that a sensor notices something in the world and turns it into a number.
- Name four sensors found at school, at home or in a phone.
- Sort everyday objects into sensors, displays and actuators.

Sense, show, decide, act

- A sensor measures
- A display shows the number
- A person or computer decides
- An actuator acts

Human sensors and real sensors: Card sort

1. Warm-up (5 min): Ask pupils to close their eyes for 30 seconds and notice everything they can: sounds, warmth, the chair, smells.
2. Bowls test (10 min): A volunteer holds one hand in the cold bowl and one in the warm bowl for a minute, then puts both in the middle bowl.
3. Teach (8 min): Read the thermometer in the middle bowl, or explain it would give one number whoever held it.
4. Card sort (12 min): Groups sort the 12 cards into three piles: sensors, displays and actuators.
5. Wrap-up (5 min): Each group shares one card they argued about.

Fill the bowls just before class; the warm water should feel comfortable, not hot. Draw 12 cards per group: thermometer, solar lamp sensor, soil probe, microphone, camera; TV screen, scale dial, scoreboard; pump, fan motor, buzzer, loudspeaker.

Let’s discuss

1. What did your ears, skin and nose notice just now?
2. So why does each hand tell a different story?
3. Does this card notice something, show something or do something?

Exit ticket

- One sensor + what it senses + where I saw it.